



GUI DESIGN · SCREEN GALLERY

Spatial · Spectral · Radiometric Sensor Modeling GUI

A first-principles electro-optical sensor performance model — propagating spectral radiance through a seven-stage signal chain from source to output.

01 Source

02 Atmosphere

03 Platform

04 Optics

05 Detector

06 Readout

07 Performance

RADIANT v1.0.0rc3 · PySide6 / Qt6 layout study

Workspace · Stage detail · Scripting

Generated June 2026

Workspace

The visual home — the seven-stage signal chain with live health indicators, parameter tree, visualization canvas, and per-stage drill-down.


RADIANT
Mid-fi · Qt6 PySide6
mock

Workspace
Scripting

Layout A D

leo_mwir_clear.yaml — RADIANT v1.0.0
#7f3a · git b4e0c1

File Edit View Run Tools Help

1 SOURCE
Source
L_src(λ)

2 ATM
Atmosphere
 τ , L_path, L_atm

3 PLATFORM
Platform
smear · jitter

4 OPTICS
Optics
PSF · MTF · τ_{opt}

5 DETECTOR
Detector
QE · noise · dark

6 READOUT
Readout
TDI · CDS · ADC

7 PERF
Performance
SNR · NEDT · NIIRS

SNR
47.3
▲ 2.1 since open

NEDT
23 mK
target ≤ 30

NIIRS
5.4
GIQES

GSD
3.6 m
@ nadir

MTF @ N_NYQ
0.42
RER 0.28

REGIME
extended
auto

F4
Validate

Sweep

Evaluated
F5

PARAMETERS
expand all clear

Search any parameter...

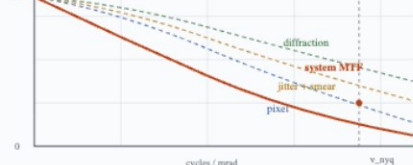
sensor.optics
aperture_diameter 0.30 m
focal_length 1.20 m
f_number 4.0
obscuration_ratio 0.33
wfe_rms ±σ 0.07 λ
temperature 280 K

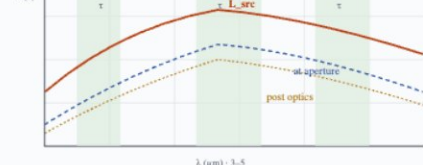
sensor.detector
material HgCdTe
pixel_pitch 18.0 μ m
cutoff_wavelength 5.0 μ m
temperature 80 K
dark_current 0.12 e⁻/s

sensor.readout
integration_time 5.0 ms
gain 1.0 e⁻/DN
adc_bits 14
cds on

scenario.atmosphere
scenario.geometry
scenario.target
scenario.background
scenario.platform

VISUALIZATION · ATMOSPHERE
reset view export PNG

System MTF
v_nyq = 27.8 cyc/mr...


Spectral radiance
500 λ pts · 3-5 ...


ACTIONS

Evaluate chain F5

Parameter sweep... F5

Monte Carlo...

Compare with...

Open YAML...

Save session

RECENTS

leo_mwir_clear.yaml

geo_swir_hazy.yaml

baseline_lwir.yaml

sarah_sweep_17.yaml

raj_handoff_3.yaml

REGIME

extended point sub-pixel

finalized in OpticsStage (C7 in master arch)

PROVENANCE

run · 2026-04-18T14:30Z

sha · 7f3a9e2b

git · b4e0c1 (clean)

py · 3.12.2

radiant · 1.0.0rc3

Spectral MTF Noise Budget Sweep Var Explorer YAML Console

signal: 12,450 e⁻ · σ: 263 e⁻ · SNR: 47.3

photon_shot 111.6

dark_current_shot 89.2

read_noise 26.0

78%

62%

5.0%

Computes $L_src(\lambda)$. Target descriptor, spec-form picker (S1–S12), illumination and line-of-sight geometry, with a live validation rail.



Atmosphere

Propagates source radiance — transmittance $\tau(\lambda)$, path radiance $L_{\text{path}}(\lambda)$, and self-emission $L_{\text{atm}}(\lambda)$ via MODTRAN tape7.


RADIANT
Stage detail · click any chain node · 1-7
hotkeys

Workspace
Workspace · Stage detail
Scripting

1 2 3 4 5 6 7

leo_mwir_clear.yaml — RADIANT v1.0.0
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signal chain / 2. Atmosphere

Atmosphere · τ , L_{path} , L_{atm}

Propagates source radiance through the atmosphere. Produces spectral transmittance $\tau(\lambda)$, upwelling path radiance $L_{\text{path}}(\lambda)$, and atmospheric self-emission $L_{\text{atm}}(\lambda)$. Uses MODTRAN tape7 as primary model, Beer-Lambert as fallback.

model · MODTRAN 6
tape7 · mls_clear_500km.t7
altitude · 500 km
zenith · 42°

ACTIVE SCENARIO MODTRAN 6 · mid-lat summer · 500 km · clear · vis 23 km · slant 675 km

archetypes... load preset...

ATMOSPHERE · DESCRIPTOR

MODEL

Engine MODTRAN 6
tape7 file mls_clear...
profile mid-lat summer
aerosol rural

CONDITIONS

visibility 23 km
RH 45 %
CO₂ 420 ppm
H₂O scale 1.0 x

GEOMETRY

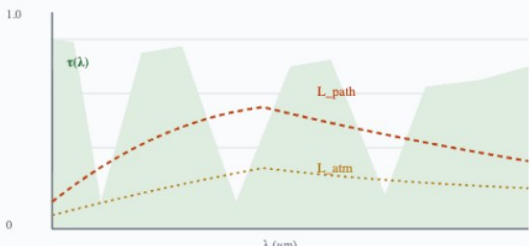
altitude 500 km
slant range 675 km
zenith θ_o 42 °
earth model WGS84

TURBULENCE

enabled false
Cn² m^{-2/3}

τ , L_{path} , L_{atm} Band summary Slant geometry Provenance MODTRAN tape7 · 1001 λ pts · 3.0–5.0 μm

Atmosphere output · $\tau(\lambda)$, $L_{\text{path}}(\lambda)$, L_{a} ...
band 3–5 μm · cache hit · 3 _



RADIATIVE TRANSFER

$$L_{\text{app}}(\lambda) = \tau \cdot L_{\text{src}} + L_{\text{path}} + L_{\text{atm}}$$
 τ · transmittance
 L_{path} · upwelling path rad.
 L_{atm} · self-emission
ref: MODTRAN 6 User Manual §4

BAND-INTEGRATED SUMMARY

BAND	τ	L_{PATH}	L_{ATM}
3.4–4.1 μm	0.82	0.41	0.08
4.1–4.8 μm	0.71	0.58	0.12
4.8–5.0 μm	0.63	0.71	0.18

VALIDATION

VALIDATION

0 err 1 warn 2 info

WARN · A2.1

tape7 grid (1001 pts @ 1 nm) does not match upstream $\epsilon(\lambda)$ grid (500 pts @ 4 nm). Linear interpolation at the boundary.
→ atm.modtran_file

INFO · A2.2

Aerosol profile = rural · visibility 23 km. Marine/urban variants available in materials/aerosols.
→ atm.aerosol

INFO · A2.3

Turbulence disabled — Cn² profile not evaluated. Enable for <3 km horizontal paths, turbulence.enabled

DESCRIPTORS

EMITTED

ATMOSPHEREDESCRIPTOR

model MODTRAN...
tape7 mls_clear_500km · 2.3 _
sha 4a7c...
t (3–5 0...
uppath 0.57 W/m²/...
L(atm) 0.13 W/m²/...
(band)

PATHGEOMETRY

airmass 1...
slant 675 _
R0000 100 _
earth WGS...
model

RADIANT — Screen Gallery

03

Platform

Line-of-sight motion — along-track smear, Gaussian jitter, and TDI mis-registration as MTF multipliers.


RADIANT
Stage detail · click any chain node · 1-7
hotkeys

Workspace
Workspace · Stage detail
Scripting

1 2 3 4 5 6 7

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signal chain / 3. Platform

Platform · smear · jitter

Applies platform/line-of-sight motion effects: along-track **smear** from ground speed × integration time, Gaussian **jitter** from LOS stability, TDI mis-registration. All are MTF multipliers (no radiometric change).

jitter 3 μrad > spec 2 μrad
ground speed · 7.5 km/s
int time · 5 ms
TDI · 8 stages

ACTIVE SCENARIO LEO 500 km · v_g 7.5 km/s · t_int 5 ms · TDI 8 stages · jitter spec 2 μrad

archetypes... load preset...

PLATFORM · DESCRIPTOR

ORBIT / MOTION
P1 LEO
ground speed v_g 7.5 km/s
altitude 500 km
tilt 0 °

STABILIZATION
over spec
jitter_rms 3.0 μrad
jitter spec 2.0 μrad
drift @ 1Hz 0.5 μrad/s

SCAN / TDI
S2 along-track
TDI stages 8
integration t 5.0 ms
TDI mis-reg 0.05 px

LOS motion
Platform MTF
Equations
Provenance
px = 18 μm · IFOV 15 μrad

LOS motion · smear + jitter over integration wi...
v_g 7.5 km/s · t_int 5 ms

ideal ⊥
jitter σ 3 μrad Δ
smear 3 μrad/5ms

ground-track
along-track →

Platform M...
smear · jitter · T_

MOTION EQUATIONS

$$MTF_{smear}(v) = \text{sinc}(v \cdot v \cdot t_{int})$$

$$MTF_{jitter}(v) = \exp(-2\pi^2 \sigma^2 v^2)$$

$$MTF_{tdi}(v) = \text{sinc}(v \cdot \Delta_{mis})$$
refs: Holst 2011 · Fiete 2010

VALIDATION

VALIDATION
2

0 err 1 warn 1 info

WARN · P3.1
jitter_rms (3.0 μrad) exceeds spec (2.0 μrad). MTF_jitter @ v_nyq: 0.96 → 0.91.
Review stabilization budget.
→ sensor.stabilization.jitter_rms

INFO · P3.2
TDI mis-registration (0.05 px) within tolerance. MTF_tdi(v_nyq) = 0.998.
→ sensor.scan.tdi_mis_reg

DESCRIPTORS EMITTED
2

PLATFORMDESCRIPTOR
P1

v_ground 7.5 km...
altitude 500 ...
jitter_rms 3.0 μr...
jitter_spec 2.0 μr...

MTFCONTRIBUTION
x3

smear 0...
jitter 0...
TDI mis- 0.9...
C00@bined 0...

Δ JITTER OUT OF SPEC

RMS jitter (3.0 μrad) exceeds system spec (2.0 μrad). Impact: MTF_jitter @ v_nyq drops 0.96 → 0.91. Action: review stabilization budget in sensor.stabilization.jitter_rms.

RADIANT — Screen Gallery

04

Optics

Diffraction PSF, MTF terms and encircled energy; warm-optics self-emission; filter bandpass; regime finalized (C7).


RADIANT
 Stage detail · click any chain node · 1-7
 hotkeys

Workspace

Workspace · Stage detail

Scripting

leo_mwir_clear.yaml — RADIANT v1.0.0

#7f3a · git b4e0c1

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2 ATM Atmosphere τ, L_path, L_atm

3 PLATFORM Platform smear · jitter

4 OPTICS Optics PSF · MTF · τ_opt

5 DETECTOR Detector QE · noise · dark

6 READOUT Readout TDI · CDS · ADC

7 PERF Performance SNR · NEDT · NIIRS

signal chain / 4. Optics

Optics · PSF · MTF · τ_opt
 Transforms source + atmosphere radiance into a focal-plane image. Applies throughput τ_opt, filter bandpass, diffraction PSF, WFE, obscuration, warm-optics self-emission, and finalizes the target regime.

D · 0.30 m · f/# 4.0
 WFE · 0.07 λ (Strehl 0.82)
 T_opt · 280 K
 EE_box · 0.57

ACTIVE SCENARIO

D = 0.30 m · f/4.0 · 4.2 μm · BP 1.2 μm · WFE 0.07 λ · T_opt 280 K

archetypes... load preset...

OPTICS · DESCRIPTOR

APERTURE

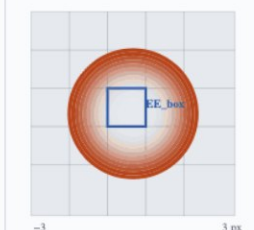
A1
 D = 0.30 m
 focal length 1.20 m
 f/# 4.0
 obscuration 0.33
 WFE_rms ±σ 0.07 λ
 T_opt 280 K
 throughput τ_opt 0.75

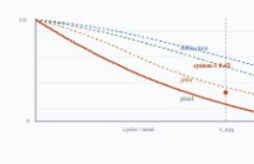
FILTER

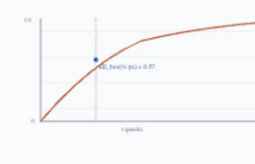
F2 BP
 type bandpass
 center λ_c 4.2 μm
 FWHM 1.2 μm
 OOB rejection 1e-4

COLD STOP

on
 enabled true
 f/# match 4.0

PSF(x...)
 λ=4.2 μm · Airy + W...


MTF ter...
 diffraction · WFE · defoc...


Encircled ene...
 EE(...)


DIFFRACTION + WFE

$$MTF_{diff}(v) = (2/\pi) [\arcsin(\beta) - \beta \sqrt{1-\beta^2}]$$

$$S = \exp(-(2\pi \cdot \sigma_{wfe})^2) \text{ \# Marechal}$$

$$\beta = v \cdot \lambda \cdot f/\#$$

WARM-OPTICS SELF-EMISSION

$$L_{opt}(\lambda) = (1 - \tau_{opt}) \cdot L_{BB}(\lambda, T_{opt})$$
 important > 3 μm; dominant for LWIR.

REGIME · FINALIZED HERE (C7)

extended ← final
 point
 sub-pixel

 PSF FWHM = 1.8 px · target footprint = 9 px → extended-scene equations downstream.

VALIDATION

0 err 0 warn 3 info

INFO · 04.1
 Strehl 0.82 (Marechal). PSF FWHM = 1.8 px. Target footprint 9 px → regime FINAL = extended (C7).
 → optics.regime

INFO · 04.2
 Cold stop on, f/# matched. Stray-light contribution from outside cold-stop modeled at 0.
 → optics.coldstop

INFO · 04.3
 Warm-optics self-emission peaks 4.4 μm at T_opt = 280 K. Contributes 0.18 W/m²/sr in band.
 → optics.T_opt

DESCRIPTORS EMITTED

OPTICSDESCRIPTOR

A1 · F2
 D 0.30 m
 f/# 4 m
 WFE 0.07 m
 Strehl 0 m
 EE_box 0 m
 regime extended (FINAL)

FILTERDESCRIPTOR

F2
 type bandpa...
 λ_c 4.2 m
 FWHM 1.2 m
 OOB 1e...

Detector

Converts radiance to electrons — $QE(\lambda)$, Rule-07 dark current, eight-term noise budget, and the spectral-integration boundary (C5).


RADIANT
Stage detail · click any chain node · 1-7
hotkeys

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Workspace · Stage detail
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leo_mwir_clear.yaml — RADIANT v1.0.0

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QE · noise · dark

6 READOUT
Readout
TDI · CDS · ADC

7 PERF
Performance
SNR · NEDT · NIIRS


signal chain / 5. Detector
Detector · QE · noise · dark
Converts incident radiance to electrons. Applies $QE(\lambda)$, dark current (Rule 07), pixel-aperture MTF, IPC/charge diffusion. This is also where **Spectral Integration** collapses the spectral dimension into per-pixel scalars (C5, exactly once).

material · HgCdTe
pitch · 18 μ m · λ_c 5.0 μ m
T_det · 80 K · dark 0.12 e⁻/s
8 noise terms active

ACTIVE SCENARIO HgCdTe · 18 μ m pitch · λ_c = 5.0 μ m · T_det 80 K · C5 spectral integration

archetypes... load preset...

DETECTOR · DESCRIPTOR

FPA

D2
HgCdTe

material HgCdTe
pixel pitch 18.0 μ m
 λ_c 5.0 μ m
T_det 80 K
fill factor 0.95
well capacity 2e6 e⁻
QE peak 0.78

CROSSTALK

X1
IPC

IPC 0.02
charge diffusion 1.2 μ m

NON-UNIFORMITY

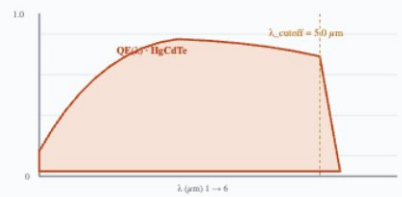
N1
NUC

DSNU 0.05 %
PRNU 0.8 %
NUC residual 0.2 %

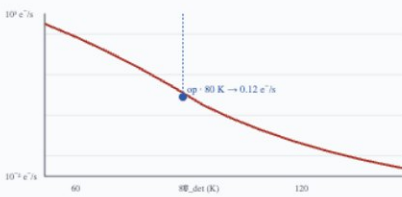
QE & Dark
Noise budget
Spectral Integration
Provenance

8 noise terms · 12,450 e⁻ signal

QE...
HgCdTe · λ_c 5.0 μ m



Dark current v...
Rule 07 · 3_d(λ)



Noise budget · this st...
e⁻ R...

TERM	E ⁻	CONTRIBUTION	%
photon_shot	111.6		18.0
dark_current_shot	89.2		11.5
ipc_crosstalk	8.1		0.09
prnu_residual	7.3		0.08
dsnu_residual	4.2		0.03
nuc_residual	3.5		0.02
glow	2.0		0.01
persistence	1.2		0.00

SPECTRAL INTEGRATION (C5) · HAPPENS HERE, EXACTLY ONCE

$$n_e = (A \cdot t / h\nu) \cdot \int L(\lambda) \cdot \tau(\lambda) \cdot QE(\lambda) \cdot BP(\lambda) \cdot d\lambda$$
EE box applied here for point/sub-pixel regimes only (C6)

VALIDATION

3

0 err
1 warn
2 info

WARN · D5.1
Photon-shot dominates noise (78%) — system is photon-limited at this scenario. Increasing aperture or t_int gives best SNR returns; reducing dark current has minor effect on detector.noise

INFO · D5.2
C5 boundary: $\int L(\lambda) \cdot \tau(\lambda) \cdot QE(\lambda) \cdot BP(\lambda) \cdot d\lambda$ collapses spectral arrays to scalar n_e. Stages downstream operate on scalars only. → detector.spectral_integration

INFO · D5.3
Rule-07 dark = 0.12 e⁻/s @ T=80K. xT_int (5 ms) = 0.0006 e⁻/pixel. Negligible at this T. → detector.T_det

DESCRIPTORS
EMITTED

2

DETECTORDESCRIPTOR

D2

material HgCdTe
pitch 18 μ m
T_det 80 K
QE 0.78
dark 0.12 e⁻/s
n_signal 12,450

NOISETERMS

x8

photon_shot 111.6
dark_shot 89.2
IPC 8.1
PRNU + 11.5

Readout

TDI / CDS timing and ADC transfer — read, 1/f, kTC and quantization noise, producing digitized counts per pixel.


RADIANT
Stage detail · click any chain node · 1-7
hotkeys

Workspace · **Workspace · Stage detail** · Scripting

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signal chain / 6. Readout
Readout · TDI · CDS · ADC

TDI · 8 stages
CDS · on
ADC · 14-bit · 1.0 e⁻/DN
frame · 141 Hz

ACTIVE SCENARIO **TDI 8 stages** · CDS on · 14-bit ADC · gain 1.0 e⁻/DN · frame 141 Hz

archetypes... load preset...

READOUT · DESCRIPTOR

READOUT CHAIN

integration t 5.0 ms
CDS on
Fowler N 1
gain 1.0 e⁻/DN
ADC bits 14
coadds 1

TDI

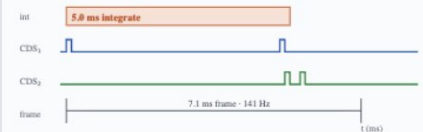
stages 8
clock jitter 0.1 ns

READOUT NOISE

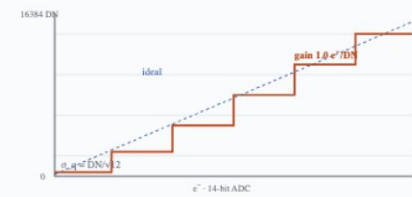
read_noise 25 e⁻
1/f 12 e⁻
kTC 0 e⁻

Timing · ADC Noise · readout Equations Provenance

Readout timing · ...
integrate · reset · samp...



ADC trans...
14-bit · gain 1.0 e⁻/...



READOUT SCALING

$$N_{sig_tdi} = N_{sig} \cdot N_{tdi} \cdot (1 - \Delta_{mis}/px)^2$$

$$\sigma_{read_cds} = \sigma_{read} \cdot \sqrt{2} \text{ \# two reads}$$

$$\sigma_{quant} = (e^-/DN) / \sqrt{12}$$

$$N_{total} = N_{sig_tdi} \cdot n_{coadds}$$

refs: Janesick 2007 · Holst 2011

READOUT NOISE TERMS

TERM	E ⁻	CONTRIBUTION	NOTES
read_noise	25.0		CDS √2 applied
1/f noise	12.0		integration-length scaling
kTC	0.0	—	reset → cancelled by CDS
quantization	3.2		DN/√12 · 14-bit

VALIDATION

VALIDATION

0 err 0 warn 2 info

INFO · R6.1
CDS active — kTC reset noise cancelled (correlated subtraction). σ_{read} scales by √2 (two reads).
→ readout.cds

INFO · R6.2
ADC well utilization 76% (12,450 / 16,384 DN). Headroom OK; no saturation.
→ readout.adc

DESCRIPTORS EMITTED

READOUTDESCRIPTOR

CDS 0
gain 1.0 e⁻/...
ADC 14-b...
DN_{out} 12,4...
σ_{total} 263 ...

NOISETERMS

read (CDS) 35.4 ...
1/f 12 ...
kTC cancell...
quantization 3.2 ...

Evaluated 0.23s Readout applied · 12,450 e⁻ · σ_{total} 263 e⁻ · DN 12,450 · 14-bit

provenance #7f3a · radiant 1.0.0rc3

RADIANT — Screen Gallery

07

Performance

SNR, NEDT, NIIRS (GIQE5/IIRS), GSD, MTF @ Nyquist, RER and detection range, with a pass/fail matrix and provenance manifest.


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signal chain / 7. Performance
Performance · SNR · NEDT · NIIRS
Computes the final radiometric performance metrics from the evaluated chain: **SNR, NEDT, NIIRS** (GIQE5 / IIRS), **GSD, MTF @ v_nyq, RER, detection range**. This stage is read-only from ChainState — no new physics, only metric aggregation.

evaluated · 0.23 s
500 λ pts · 8 noise terms
regime · **extended**
provenance · #7f3a

ACTIVE SCENARIO 6 metrics · 5 PASS · 1 WATCH · GIQE5 · run #7f3a

archetypes... load preset...

PERFORMANCE METRICS · EVALUATED

SNR
47.3

NEDT
23 mK

NIIRS (GIQE5)
5.4

GSD
3.6 m

MTF @ N_NYQ
0.42

RER
0.28

PERFORMANCE · DESCRIPTOR

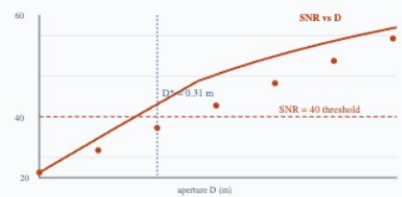
TARGETS
SNR threshold 40
NEDT threshold 30 mK
NIIRS target 5.0

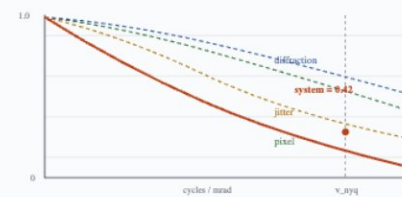
MODELS
NIIRS model M1 GIQE5
detection johnson-nvl
EE box r 0.5 px

RANGE
P_detect 0.80
P_recognize 0.50
P_identify 0.30

SNR · sweep System MTF Equations Targets

GIQE5 · 8 noise terms · 500 λ

SNR vs aperture (last sw...
D* = 0.31 m @ SNR=...


System ...
v_nyq · R_...


SNR · NEDT
 $SNR = N_{sig} / \sqrt{\Sigma \sigma_i^2}$
 $NEDT = \sigma_{total} / (\partial N / \partial T)$
 $\partial N / \partial T$ · via finite diff on source at ±0.5 K

NIIRS · GIQE5 (EO), IIRS (IR)
 $NIIRS = 9.57 + c \cdot \log(GSD) + \dots$
· MTF @ v_nyq · RER · SNR
refs: IRARS GIQE5 · Leachtenauer 1997

DETECTION RANGE
 $R_{det} = f(char\ size, MRC, SNR, P_{det})$
target 1m @ P=0.8 → 4.2 km

RER
 $RER = \frac{1}{2} \cdot |ESF(+\frac{1}{2}) - ESF(-\frac{1}{2})|$
proxy for ground-sampled resolution

VALIDATION

VALIDATION
5 pass 1 watch 1 info

WARN · V7.1
jitter_rms (3.0 μrad) exceeds spec (2.0 μrad). MTF_jitter contribution acceptable but should be tracked. Source: PlatformStage.
→ sensor.stabilization.jitter_rms

INFO · V7.2
All 6 chain stages evaluated cleanly. 8 noise terms summed, 7 MTF terms multiplied.
Provenance hash #7f3a9e2b.
→ perf.evaluation

DESCRIPTORS
EMITTED

CHAINRESULT #7f3a

SNR 47...
NEDT 23...
NIIRS 5...
GSD 3.6...
MTF @ 0...
RERYQ 0...
range @ 4.2...
P=0.8

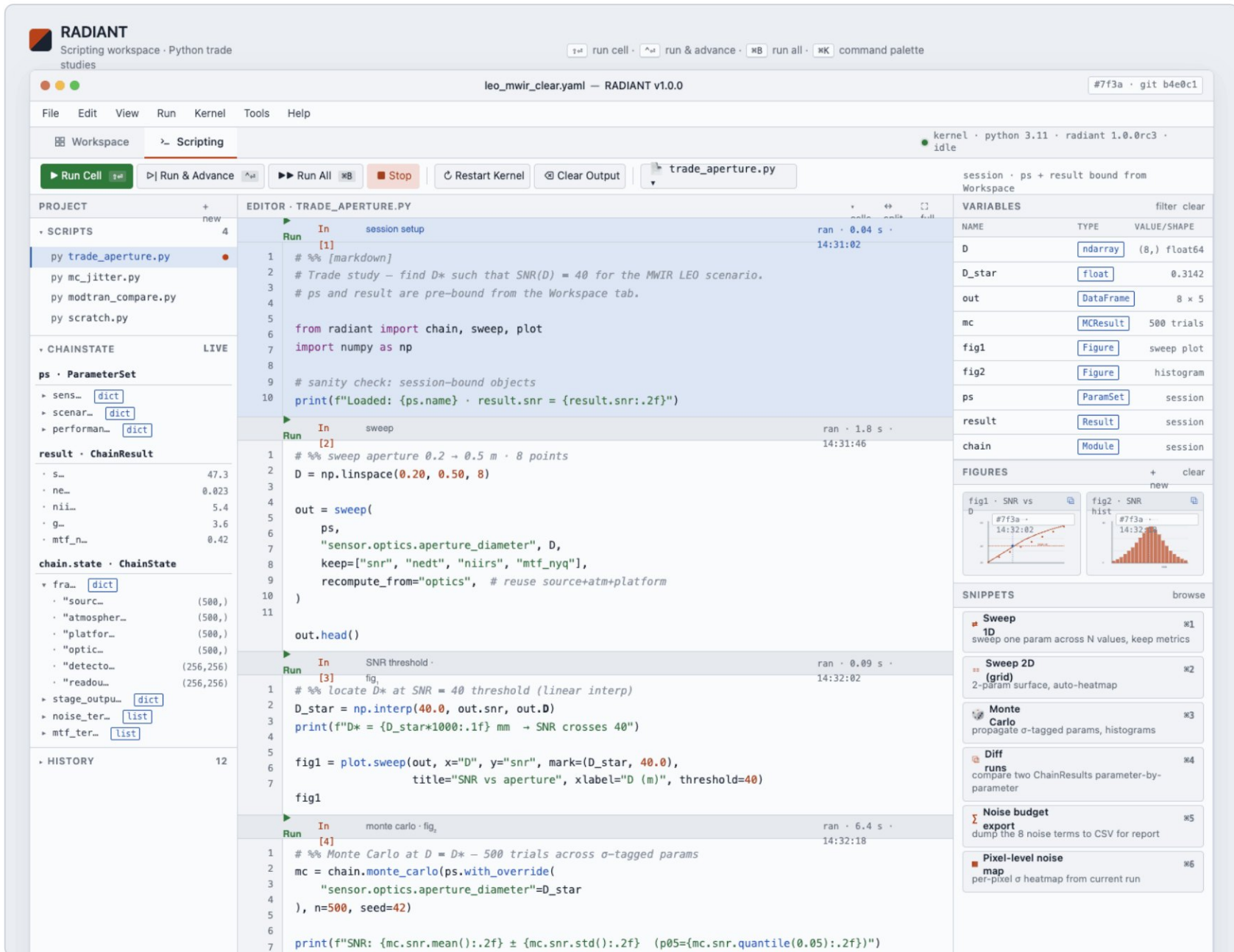
PROVENANCE attached

RADIANT — Screen Gallery

08

Scripting

The Python trade-study workspace — cell editor, live ChainState explorer, figures dock, and console, sharing the visual Workspace backend.



RADIANT
Scripting workspace · Python trade studies

leo_mwir_clear.yaml — RADIANT v1.0.0 #7f3a · git b4e0c1

File Edit View Run Kernel Tools Help

Workspace Scripting

Run Cell Run & Advance Run All Stop Restart Kernel Clear Output trade_aperture.py

session · ps + result bound from Workspace

PROJECT

- SCRIPTS
 - py trade_aperture.py
 - py mc_jitter.py
 - py modtran_compare.py
 - py scratch.py
- CHAINSTATE LIVE
 - ps · ParameterSet
 - sens... dict
 - scenar... dict
 - performan... dict
 - result · ChainResult
 - s... 47.3
 - ne... 0.023
 - nii... 5.4
 - g... 3.6
 - mtf_n... 0.42
 - chain.state · ChainState
 - fra... dict
 - "sourc... (500,)
 - "atmosph... (500,)
 - "platfor... (500,)
 - "optic... (500,)
 - "detecto... (256,256)
 - "readou... (256,256)
 - stage_outpu... dict
 - noise_ter... list
 - mtf_ter... list
- HISTORY 12

EDITOR · TRADE_APERTURE.PY

```

1  # %% [markdown]
2  # Trade study - find D* such that SNR(D) = 40 for the MWIR LEO scenario.
3  # ps and result are pre-bound from the Workspace tab.
4
5  from radiant import chain, sweep, plot
6  import numpy as np
7
8  # sanity check: session-bound objects
9  print(f"Loaded: {ps.name} · result.snr = {result.snr:.2f}")
10
11  # %% sweep aperture 0.2 → 0.5 m · 8 points
12  D = np.linspace(0.20, 0.50, 8)
13
14  out = sweep(
15      ps,
16      "sensor.optics.aperture_diameter", D,
17      keep=["snr", "nedt", "niirs", "mtf_nyq"],
18      recompute_from="optics", # reuse source+atm+platform
19  )
20
21  out.head()
22
23  # %% locate D* at SNR = 40 threshold (linear interp)
24  D_star = np.interp(40.0, out.snr, out.D)
25  print(f"D* = {D_star*1000:.1f} mm → SNR crosses 40")
26
27  fig1 = plot.sweep(out, x="D", y="snr", mark=(D_star, 40.0),
28                  title="SNR vs aperture", xlabel="D (m)", threshold=40)
29  fig1
30
31  # %% Monte Carlo at D = D* - 500 trials across sigma-tagged params
32  mc = chain.monte_carlo(ps.with_override(
33      "sensor.optics.aperture_diameter"=D_star
34  ), n=500, seed=42)
35
36  print(f"SNR: {mc.snr.mean():.2f} ± {mc.snr.std():.2f} (p05={mc.snr.quantile(0.05):.2f})")

```

VARIABLES

NAME	TYPE	VALUE/SHAPE
D	ndarray	(8,) float64
D_star	float	0.3142
out	DataFrame	8 × 5
mc	MCResult	500 trials
fig1	Figure	sweep plot
fig2	Figure	histogram
ps	ParamSet	session
result	Result	session
chain	Module	session

FIGURES

- fig1 · SNR vs D
- fig2 · SNR hist

SNIPPETS

- Sweep 1D: sweep one param across N values, keep metrics
- Sweep 2D (grid): 2-param surface, auto-heatmap
- Monte Carlo: propagate sigma-tagged params, histograms
- Diff runs: compare two ChainResults parameter-by-parameter
- Noise budget export: dump the 8 noise terms to CSV for report
- Pixel-level noise map: per-pixel sigma heatmap from current run